

# Horse Racing Tournament Management System – Requirements

## 1. Authentication & Authorization

- Register account
  - Login
  - Logout
  - JWT Authentication
  - Role-based Authorization
  - Manage profile
  - Block/Activate account
  - Password hashing
  - Session/token management
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## 2. User Roles

- Admin
  - Horse Owner
  - Jockey
  - Race Referee
  - Spectator
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## 3. Admin Requirements

### User Management

- Create user
- Update user
- Delete user
- Block user
- View user list
- Manage roles
- Manage permissions

### Tournament Management

- Create tournament
- Update tournament
- Delete tournament
- Manage tournament status
- Create rounds
- Create races
- Schedule races
- Set race location
- Set race date/time
- Manage prize pool
- Prevent schedule conflicts

## **Registration Management**

- Approve horse registration
- Reject horse registration
- Manage pending registrations

## **Horse Management**

- View approved horses
- Assign horse racing ID
- Remove invalid horses

## **Jockey Management**

- View jockey list
- Update jockey status
- Suspend jockey

## **Referee Management**

- Assign referee to race
- View available referees
- Send assignment notifications

## **Result Management**

- Review raw results
- Publish official results
- Update leaderboard
- Calculate rewards/prize money

## **Prediction System**

- Configure prediction settings
- Configure reward rate
- Open/close prediction gate
- View prediction statistics

## **Reporting & Statistics**

- Dashboard
  - Total tournaments
  - Total horses
  - Total races
  - Top horses
  - Top jockeys
  - Win rate
  - Revenue statistics
  - Monthly race statistics
  - Export reports PDF
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# **4. Horse Owner Requirements**

## **Account**

- Register Horse Owner account
- Manage profile

## **Horse Management**

- Add horse
- Edit horse
- Delete horse
- View horse details
- Upload horse image
- Manage horse records

## **Horse Registration**

- Register horse for tournament
- View registration status

## **Jockey Invitation**

- Search jockey
- View jockey profile
- Send jockey invitation
- Manage invitations
- Cancel invitation

## **Jockey Assignment**

- Confirm official jockey
- Remove jockey from horse

## **Race Participation**

- Confirm horse participation
- View race details
- View race schedule
- Filter schedule

## **Performance Tracking**

- View horse results
  - View leaderboard
  - View reward/prize history
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# **5. Jockey Requirements**

## **Account**

- Register Jockey account
- Manage profile

## **Invitations**

- View invitations
- Accept invitation
- Reject invitation
- View invitation details

## **Race Participation**

- View assigned races
- View assigned horse details
- View personal race schedule

## **Performance**

- View race history
  - View leaderboard
  - View statistics
  - View win rate
  - View achievements
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# **6. Race Referee Requirements**

## **Race Check-in**

- View participant list
- Check horse condition
- Check jockey equipment
- Approve/Reject check-in

## **Race Monitoring**

- Monitor race progress
- Update live race information
- Update horse positions
- Record race events

## **Violation Management**

- Record violations
- Select violation type
- Add penalty
- Add violation notes

## **Result Management**

- Review finish order
- Adjust ranking based on penalties
- Submit raw results

## Reports

- Create race report
  - Add race comments
  - Sign report
  - Submit report
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# 7. Spectator Requirements

## Tournament Viewing

- View tournaments
- View tournament details
- View race schedules
- View participants

## Live Tracking

- View live races
- View live rankings
- View live events

## Leaderboard

- View tournament leaderboard
- View horse rankings
- View jockey rankings

## Prediction System

- Predict race winner
  - Predict top participants
  - Submit prediction
  - View prediction history
  - View prediction status
  - Claim rewards
  - Receive reward notifications
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## 8. Race Management Requirements

- Create race
  - Assign participants
  - Assign jockey
  - Assign referee
  - Set lane number
  - Manage race status
  - Track race progress
  - Manage race results
  - Manage violations
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## 9. Result Management Requirements

- Store finish time
  - Store rankings
  - Store penalties
  - Store reward money
  - Publish official results
  - Auto-update leaderboard
  - Auto-calculate rewards
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## 10. Notification Requirements

- Invitation notifications
  - Race schedule notifications
  - Result notifications
  - Prediction reward notifications
  - Referee assignment notifications
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## 11. Search & Filter Requirements

- Search horse
- Search jockey
- Search tournament
- Search race
- Filter schedules

- Filter leaderboard
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## 12. Upload Requirements

- Upload horse images
  - Upload jockey avatars
  - Upload tournament images
  - Upload verification documents
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## 13. Reporting Requirements

- Tournament reports
  - Race reports
  - Referee reports
  - Prediction reports
  - Revenue reports
  - Export PDF
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## 14. Non-Functional Requirements

### Security

- JWT Authentication
- Password hashing
- Authorization by role
- Secure API access
- Input validation

### Performance

- Support multiple races
- Fast leaderboard update
- Real-time race tracking

### Scalability



- Extendable architecture
- Support future betting system
- Support mobile app
- Support livestream integration

## Usability

- Responsive UI
  - Dashboard visualization
  - Easy navigation
  - Search & filter support
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